Surface Area of Prisms, Cylinders, Cones and Spheres

Name _____ Date _____ Calculator allowed Version 1

Scan to practise this topic online

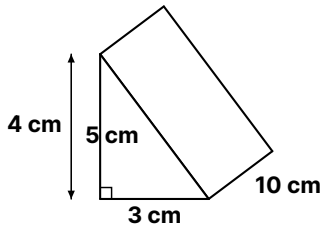
KEY WORDS surface area *(horahanga mata)* · curved surface · net *(raupapa mata)* · slant height *(teitei tītaha)*

A Triangular prisms ●○○

Remember: Think of the net: two triangles and three rectangles. Each rectangle is one side of the triangle times the length of the prism.

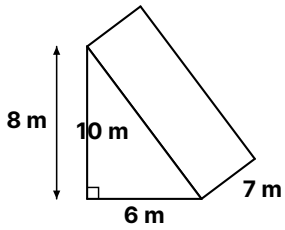
1. Find the surface area of each prism.

a)



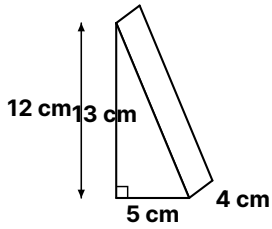
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b)



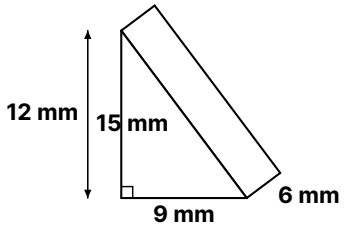
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c)



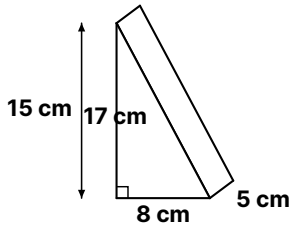
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d)



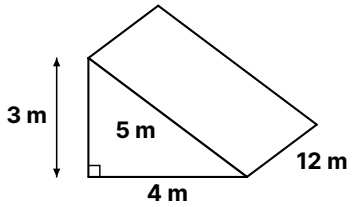
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e)



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f)



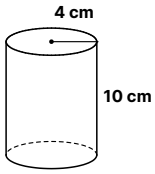
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B Cylinders ●●○

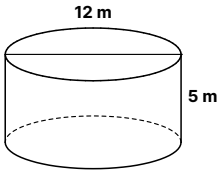
Remember: The net of a cylinder is two circles and a rectangle whose length is the circumference: SA $= 2\pi r^2 + 2\pi rh$. Give answers to 1 decimal place.

2. Find the surface area of each cylinder. In (e) and (f) find the curved surface only.

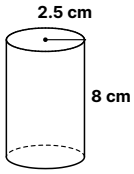
a)



b)

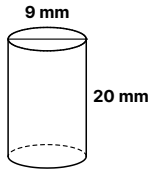


c)



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d)



e) The curved surface only of a cylinder with radius 3 cm and height 7 cm

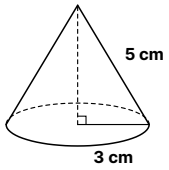
f) The curved surface only of a cylinder with diameter 10 m and height 4 m

C Cones ●●○

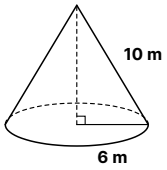
Remember: Curved surface of a cone $= \pi rl$, where l is the slant height. Add the base circle πr^2 for the total. If only the height is given, find l with Pythagoras first.

3. Find the total surface area of each cone, to 1 decimal place.

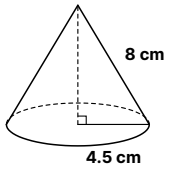
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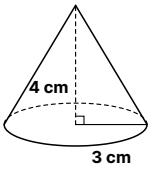
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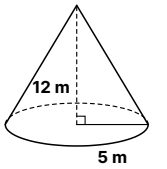
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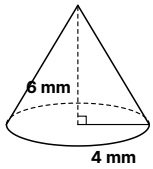
d)



e)



f)

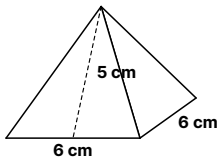


D Pyramids ●●●

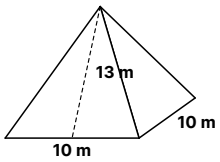
Remember: A square pyramid is a square and four identical triangles. The height of each triangle is the slant height of the face.

4. Find the surface area of each square-based pyramid.

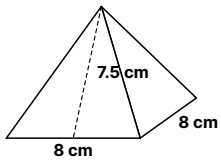
a)



b)



c)

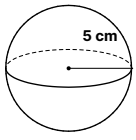


E Spheres and hemispheres ●●○

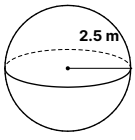
Remember: Surface area of a sphere $= 4\pi r^2$. A solid hemisphere has half of that plus its flat circular face.

5. Find the surface area, to 1 decimal place. In (d) to (f) the solid is a hemisphere (half a sphere), including its flat face.

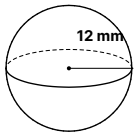
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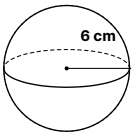
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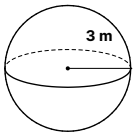
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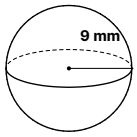
d)



e)



f)



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F Problems ●●●

6. A cylindrical water tank has a radius of 1.5 m and a height of 2 m. The whole outside, including the top and bottom, is to be painted. One litre of paint covers 9 m². How many litres are needed, to 1 decimal place?

7. An ice cream wafer is a cone (no base) with a radius of 3 cm and a slant height of 11 cm. Find the area of wafer, to 1 decimal place.

8. A ball has a radius of 11 cm. The leather for a ball costs $20 per square metre. Find the cost of the leather for one ball, to the nearest cent.

9. A tent is a triangular prism lying on one rectangular face. Its end is a right-angled triangle with sides 3 m, 4 m and 5 m (the 3 m side is on the ground), and the tent is 6 m long. Find the area of cloth, counting the two sloping faces, the two triangular ends, but not the floor.

★ Extension

10. A sphere has a surface area of 500 cm². Find its radius, to 1 decimal place.

11. A cylinder and a sphere have the same radius r, and the cylinder's height is 2r (so the sphere fits exactly inside it). Compare their curved surface areas.

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Surface Area of Prisms, Cylinders, Cones and Spheres

Name _____ Date _____ Calculator allowed Version 2

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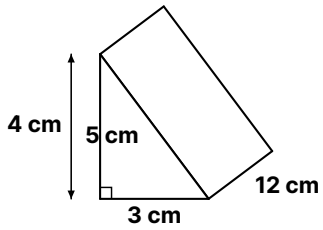
KEY WORDS surface area *(horahanga mata)* · curved surface · net *(raupapa mata)* · slant height *(teitei tītaha)*

A Triangular prisms ●○○

Remember: Think of the net: two triangles and three rectangles. Each rectangle is one side of the triangle times the length of the prism.

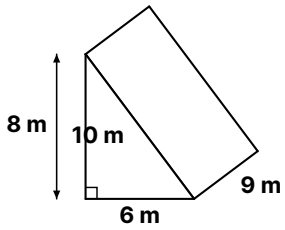
1. Find the surface area of each prism.

a)



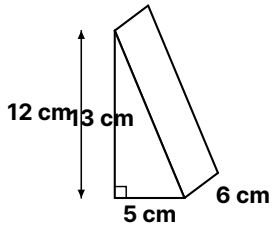
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b)



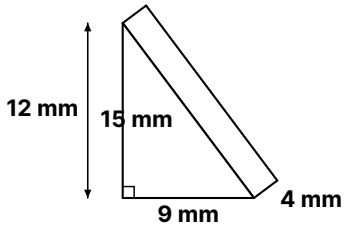
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c)



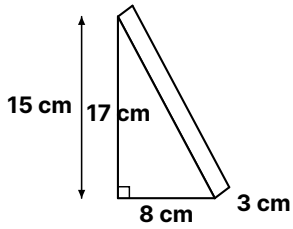
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d)



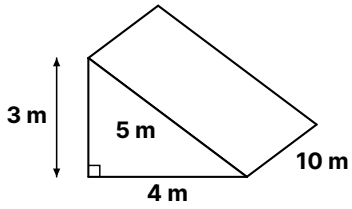
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e)



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f)



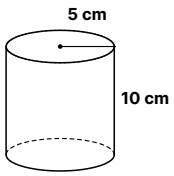
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B Cylinders ●●○

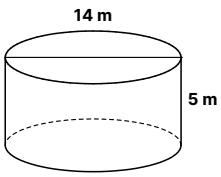
Remember: The net of a cylinder is two circles and a rectangle whose length is the circumference: SA $= 2\pi r^2 + 2\pi rh$. Give answers to 1 decimal place.

2. Find the surface area of each cylinder. In (e) and (f) find the curved surface only.

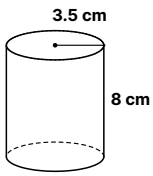
a)



b)

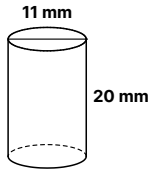


c)



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d)



e) The curved surface only of a cylinder with radius 4 cm and height 7 cm

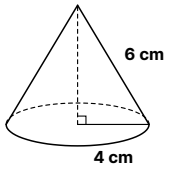
f) The curved surface only of a cylinder with diameter 12 m and height 4 m

C Cones ●●○

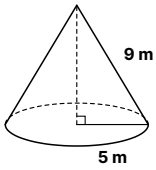
Remember: Curved surface of a cone $= \pi rl$, where l is the slant height. Add the base circle πr^2 for the total. If only the height is given, find l with Pythagoras first.

3. Find the total surface area of each cone, to 1 decimal place.

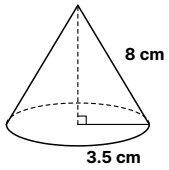
a)



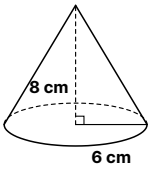
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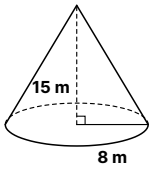
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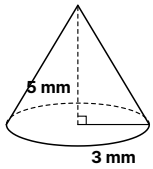
d)



e)



f)

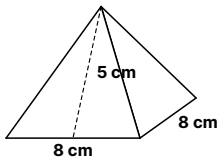


D Pyramids ●●●

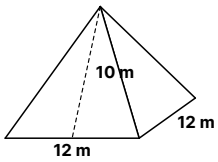
Remember: A square pyramid is a square and four identical triangles. The height of each triangle is the slant height of the face.

4. Find the surface area of each square-based pyramid.

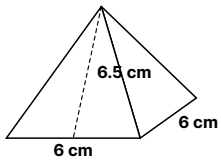
a)



b)



c)

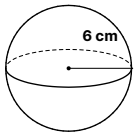


E Spheres and hemispheres ●●○

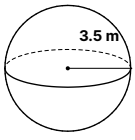
Remember: Surface area of a sphere $= 4\pi r^2$. A solid hemisphere has half of that plus its flat circular face.

5. Find the surface area, to 1 decimal place. In (d) to (f) the solid is a hemisphere (half a sphere), including its flat face.

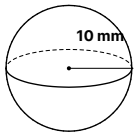
a)



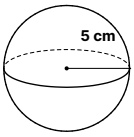
b)



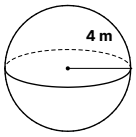
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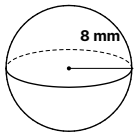
d)



e)



f)



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F Problems ●●●

6. A cylindrical water tank has a radius of 1.2 m and a height of 2.5 m. The whole outside, including the top and bottom, is to be painted. One litre of paint covers 8 m². How many litres are needed, to 1 decimal place?

7. An ice cream wafer is a cone (no base) with a radius of 3.5 cm and a slant height of 12 cm. Find the area of wafer, to 1 decimal place.

8. A ball has a radius of 12 cm. The leather for a ball costs $24 per square metre. Find the cost of the leather for one ball, to the nearest cent.

9. A tent is a triangular prism lying on one rectangular face. Its end is a right-angled triangle with sides 6 m, 8 m and 10 m (the 6 m side is on the ground), and the tent is 5 m long. Find the area of cloth, counting the two sloping faces, the two triangular ends, but not the floor.

Extension

10. A sphere has a surface area of 800 cm². Find its radius, to 1 decimal place.

11. A cylinder and a sphere have the same radius r, and the cylinder's height is 2r (so the sphere fits exactly inside it). Compare their curved surface areas.

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